MTH1003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Mathematical Modelling

Module Leader: Prof John Thuburn, Prof Andrew Gilbert

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination

SECTION A

1. (a) A ball is projected from the origin with speed V at an angle α to the horizontal. Neglecting air resistance and taking the gravitational acceleration to be g, write down expressions for the ball's horizontal and vertical coordinates x and z as functions of time t. Hence obtain the equation of the trajectory. Find the angles α that give a range equal to $V^2/2g$. (8)

   (b) Consider the following short Matlab code:

```
y = 0;
z = 0;
while z < 20
    y = y + 1;
    z = z + y*y*(-1)^y;
end
disp(z)
```

   What value will be displayed when the code is executed? Explain your answer in terms of the values taken by the variables `y` and `z`. (6)

   (c) Find all the equilibrium points of the following dynamical system:

$$\dot{x} = x(y-2), \qquad \dot{y} = (x-1)(y^2-4).$$

   Classify the stability of the fixed point at $x = 0,\ y = 0$ for the following dynamical systems:

$$\begin{array}{lll} \text{(i)} & \dot{x} = x - y, & \dot{y} = x + y; \\ \text{(ii)} & \dot{x} = 4x - y, & \dot{y} = x - 2y. \end{array}$$

   (12)

   (d) A particle with position $x(t)$ undergoes damped harmonic motion

$$\ddot{x} + 2k\dot{x} + \omega^2 x = 0,$$

   with initial conditions $x = 2,\ \dot{x} = 0$ at $t = 0$. For the two cases

$$\begin{array}{lll} \text{(i)} & k = 1, & \omega^2 = 4, \\ \text{(ii)} & k = 2, & \omega^2 = 4, \end{array}$$

   obtain the solution $x(t)$ and state what kind of damping occurs. (14)

(e) A particle of mass m, position $x > 0$, moves in one dimension under a conservative force $F(x) = 1/x^2 - 1/x$. Derive the energy equation for the particle, giving an explicit expression for the potential energy $V(x)$. Find the equilibrium position of the particle and determine whether it is stable or unstable. (10)

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SECTION B

2. (a) A particle moves in one dimension, with position given by the coordinate x, and experiences a force $F = 3.4\,\mathrm{N}$ at $x = 1\,\mathrm{m}$ and $F = -1.6\,\mathrm{N}$ at $x = 2\,\mathrm{m}$. By rearranging the linear interpolation formula, or otherwise, estimate the equilibrium position of the particle. (5)

   (b) Let $f(x)$ be an infinitely differentiable function. Write down Taylor series expansions for $f(x \pm h)$ in terms of f and its derivatives at x. Hence show that the estimate $\hat{f}(x)$ given by linear interpolation between $(x-h, f(x-h))$ and $(x+h, f(x+h))$ for the value $f(x)$ is second order accurate. (10)

   (c) Three identical smooth cylinders, radius l_0, are held together in a triangular arrangement by a closed loop of elastic string of natural length l_0, elastic modulus λ. Determine the elastic potential energy in the string. (10)

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3. A light elastic string has natural length l_0 and elastic modulus λ. A particle of mass m is attached a distance $l_0/4$ from one end of the string and an identical particle is attached a distance $l_0/4$ from the other end. The string is then stretched and its ends are fixed a distance $4l_0$ apart on a smooth horizontal table. The two particles move in one dimension in the direction parallel to the string.

   (a) Using a simple symmetry argument, or otherwise, determine the equilibrium state of the system. (4)

   (b) Let x and y be the displacements from equilibrium of the two particles. Show that the equation of motion can be written in the form

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = k^2 \begin{pmatrix} -6 & 2 \\ 2 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

   and give an explicit expression for k^2. (5)

   (c) Obtain the general solution for this system. (10)

   (d) Hence obtain the solution that satisfies the initial conditions $x = 0$, $y = 0$, $\dot{x} = 1$, $\dot{y} = 0$ at $t = 0$. (6)

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4. You are given the equation of motion in polar coordinates (r, θ) for a particle of mass m moving under a central force $\mathbf{F} = mf(r)\hat{\mathbf{r}}$:

$$(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)\hat{\boldsymbol{\theta}} = f(r)\hat{\mathbf{r}},$$

where $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ are the unit vectors in the radial and azimuthal directions, respectively.

(a) Define the angular momentum per unit mass h and show that it is conserved. (4)

(b) Show that

$$\ddot{r} - \frac{h^2}{r^3} = f(r). \qquad (*)$$

Hence show that

$$\frac{1}{2}(\dot{r})^2 + \frac{1}{2}\left(\frac{h}{r}\right)^2 + V(r)$$

is a constant of the motion, where $V(r) = \int -f(r)\,\mathrm{d}r$. (7)

(c) You are given that the equation $(*)$ may be written in the form

$$\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = -\frac{1}{h^2u^2}f\left(\frac{1}{u}\right),$$

where $u = 1/r$. Use this result to determine the form of the force $f(r)$, given that a particle is observed to follow the path

$$r = \frac{r_0}{\cos 3\theta}$$

with r_0 constant. (6)

(d) Assuming that the particle passes through the point $(r, \theta) = (r_0, 0)$, for what angles θ does the particle's radial coordinate tend to infinity? (4)

(e) Explain briefly why a force of the form found in part (c) cannot support steady circular orbits. (4)

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